Han Lu

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Education

Stanford Online High School — Stanford, CA
High School Diploma: 3.99 (Unweighted), College level coursework — *Aug. 2023 – June 2027*

MIT Beaver Works Summer Institute — Boston, MA
Basics of ASICS — *Class of 2026*

Experience

FIRST Robotics: FTC and FRC — August 2023 – Present
Captain, Hardware Lead, and Design Lead — *San Jose, CA; San Diego, CA*

- Learned team leadership, how to cooperate with team members. Led team members through year-to-year design ideals and onboarded new members, teaching them our process and platforms. Led school-wide workshops teaching people design.
- Third place at regional (x2), Earned Connect Award First Place (x2), Ventura County Regional Second Alliance.

JOWUA Taiwan — June 2026 – Present
Mechanical Engineer and Product Designer (Part Time), Student Intern — *Taipei, Taiwan*

- Worked on the ME (Mech. Eng.) team to develop Tesla accessories (e.g. J1772 to 6-20R transfer locking mechanism, portable vacuum attachments)
- Studied fabrication methods such as injection molding and fabrics
- Started company-wide 3D printer initiative for prototyping
- Ref. Steven Huang (CEO) steven.huang@jowua-life.com

3D Modeling and Printing Club — December 2023 – Present
President — *San Jose, CA*

- Advertising skills, conversation starters.
- Ran the 3D Printer incubator, helping amateurs get into 3D printing (Ender 3).
- Taught club attendees the basics of Autodesk Fusion 360, including CAD (computer aided design) and basic CAM (computer assisted manufacturing)

Projects

Publication: Assessment of Rule-based and RL-based ALP Solutions | *Python, PPO, Optuna, Gym*

- Paper: Assessment of Rule-Based and Reinforcement Learning Control Methods with Evaluation of Reward Parameters in Terminal Space: Empirical Comparison Using ADS-B Data (https://doi.org/10.13140/RG.2.2.20163.16160)
- Implemented FAST-style FCFS rule-based algorithm and PPO-based RL algorithms using ADS-B data from OpenSky
- Utilized Bayesian optimization reward tuning to evaluate human controller goals

BWSI Capstone: CNN Accelerator using Matrix Multipliers | *SystemVerilog, OpenLane*

- Wrote SV to create a convolutional neural network layer that takes in arbitrary NxN matrices with arbitrary size kernels.
- Ran through proper RTL to GDS flow, including testbenches, routing, DRC, LVS, antenna checks, and layout creation.

Science Bowl Question Analysis | *C++, Python, Vector Embedding*

- Created a script to download and parse official National Science Bowl (NSB)
- Processed these texts using vector embedding using SentenceTransformer
- Used K-means clustering to determine optimal concepts for each question category

Technical Skills

Languages: Java, C++, MATLAB, Verilog, SystemVerilog, Python
Hardware: Autodesk Fusion Design, Autodesk Fusion CAM, Autodesk Inventor, KiCAD, OrcaSlicer/Bambu Studio
Tools and Others: Git, VS Code, Android Studio